Exercise 3-2: Find the angle θ_{AB} between vectors **A** and **B** of Example 3-1 from the cross product between them.

Answer: $\theta_{AB} = 145.1°$. (See EM.)

Exercise 3-3: Find the angle between vector **B** of Example 3-1 and the z axis.

Answer: 101.1°. (See EM.)

Exercise 3-4: Vectors **A** and **B** lie in the y–z plane and both have the same magnitude of 2 (**Fig. E3.4**). Determine (a) $\mathbf{A}\cdot\mathbf{B}$ and (b) $\mathbf{A}\times\mathbf{B}$.

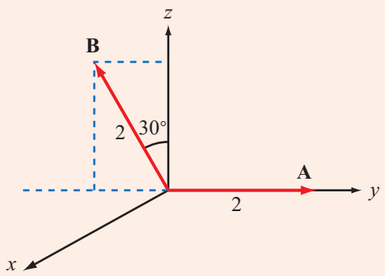


Figure E3.4

Answer: (a) $\mathbf{A}\cdot\mathbf{B} = -2$; (b) $\mathbf{A}\times\mathbf{B} = \hat{\mathbf{x}}3.46$. (See EM.)

Exercise 3-5: If $\mathbf{A}\cdot\mathbf{B} = \mathbf{A}\cdot\mathbf{C}$, does it follow that $\mathbf{B} = \mathbf{C}$?

Answer: No. (See EM.)

3-1.5 Scalar and Vector Triple Products

When three vectors are multiplied, not all combinations of dot and cross products are meaningful. For example, the product

$$\mathbf{A}\times(\mathbf{B}\cdot\mathbf{C})$$

does not make sense because $\mathbf{B}\cdot\mathbf{C}$ is a scalar, and the cross product of the vector **A** with a scalar is not defined under the rules of vector algebra. Other than the product of the form $\mathbf{A}(\mathbf{B}\cdot\mathbf{C})$, the only two meaningful products of three vectors are the scalar triple product and the vector triple product.

Scalar Triple Product

The dot product of a vector with the cross product of two other vectors is called a scalar triple product, so named because the result is a scalar. A scalar triple product obeys the cyclic order:

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}). \tag{3.29}$$

The equalities hold as long as the cyclic order ($ABCABC\ldots$) is preserved. The scalar triple product of vectors $\mathbf{A} = (A_x, A_y, A_z)$, $\mathbf{B} = (B_x, B_y, B_z)$, and $\mathbf{C} = (C_x, C_y, C_z)$ can be expressed in the form of a 3×3 determinant:

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \tag{3.30}$$

The validity of Eqs. (3.29) and (3.30) can be verified by expanding **A**, **B**, and **C** in component form and carrying out the multiplications.

Vector Triple Product

The vector triple product involves the cross product of a vector with the cross product of two others, such as

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}). \tag{3.31}$$

Since each cross product yields a vector, the result of a vector triple product is also a vector. The vector triple product does not obey the associative law. That is,

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) \neq (\mathbf{A}\times\mathbf{B})\times\mathbf{C}, \tag{3.32}$$

which means that it is important to specify which cross multiplication is to be performed first. By expanding the vectors **A**, **B**, and **C** in component form, it can be shown that

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B}), \tag{3.33}$$

which is known as the "bac-cab" rule.

Example 3-3: Vector Triple Product

Given $\mathbf{A} = \hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}2$, $\mathbf{B} = \hat{\mathbf{y}} + \hat{\mathbf{z}}$, and $\mathbf{C} = -\hat{\mathbf{x}}2 + \hat{\mathbf{z}}3$, find $(\mathbf{A}\times\mathbf{B})\times\mathbf{C}$ and compare it with $\mathbf{A}\times(\mathbf{B}\times\mathbf{C})$.

Solution:

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{\mathbf{x}}3 - \hat{\mathbf{y}} + \hat{\mathbf{z}}$$

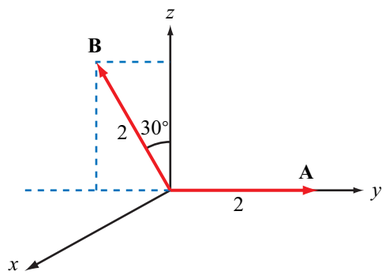
练习 3-2：根据例 3-1 中的向量 A 和 B 之间的叉积求出它们之间的角度 θ_{AB}。

Answer: $\theta_{AB} = 145.1°$. (See EM.)

练习 3-3：求例 3-1 中的向量 B 与 z 轴之间的角度。

答案：101.1°。（参见EM。）

练习 3-4：向量 A 和 B 位于 y-z 平面上，并且大小相同，均为 2（图 E3.4）。确定 (a) A · B 和 (b) A× B。



图E3.4

答案：(a) A · B = −2； (b) A× B = x̂ 3.46。 （参见EM。）

练习3-5：如果A · B = A · C，是否可以得出B = C?

答案：不。（参见EM。）

3-1.5 标量和向量的三重积

当三个向量相乘时，并非所有点积和叉积的组合都有意义。例如，产品

$$\mathbf{A}\times(\mathbf{B}\cdot\mathbf{C})$$

没有意义，因为 B ·C 是标量，并且向量 A 与标量的叉积没有根据向量代数规则定义。除了 A(B · C) 形式的乘积之外，三个向量的唯一两个有意义的乘积是标量三重乘积和向量三重乘积。

标量三重积

一个向量与另外两个向量的叉积的点积称为标量三重积，如此命名是因为

结果是一个标量。标量三重积遵循循环顺序：

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}). \tag{3.29}$$

只要保留循环顺序 (ABCABC …)，这些等式就成立。向量 A = (Ax、 Ay、 Az)、B = (Bx、 By、 Bz) 和 C = (Cx、 Cy、 Cz) 的标量三重积可以表示为 3 × 3 行列式的形式：

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}. \tag{3.30}$$

方程的有效性。 (3.29)和(3.30)可以通过将A、B和C以分量形式展开并进行乘法来验证。

向量三重积

向量三重积涉及一个向量与另外两个向量的叉积，例如

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}). \tag{3.31}$$

由于每个叉积都会产生一个向量，因此向量三重积的结果也是一个向量。向量三重积不遵守结合律。那是，

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) \neq (\mathbf{A}\times\mathbf{B})\times\mathbf{C}, \tag{3.32}$$

这意味着指定首先执行哪个叉乘非常重要。将向量A、B、C展开分量形式，可以证明：

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B}), \tag{3.33}$$

这就是所谓的 "bac-cab" 规则。

例 3-3：向量三重积

给定 A = x̂ − ŷ + ẑ2、B = ŷ + ẑ 和 C = −x̂2 + ẑ3，找到 (A× B)× C 并将其与 A× (B × C) 进行比较。

解决方案：

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{\mathbf{x}}3 - \hat{\mathbf{y}} + \hat{\mathbf{z}}$$

and

$$(\mathbf{A}\times\mathbf{B})\times\mathbf{C}=\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ -3 & -1 & 1\\ -2 & 0 & 3\end{vmatrix}=-\hat{\mathbf{x}}3+\hat{\mathbf{y}}7-\hat{\mathbf{z}}2.$$

A similar procedure gives $\mathbf{A}\times(\mathbf{B}\times\mathbf{C})=\hat{\mathbf{x}}2+\hat{\mathbf{y}}4+\hat{\mathbf{z}}$. The fact that the results of two vector triple products are different demonstrates the inequality stated in Eq. (3.32).

Concept Question 3-1: When are two vectors equal and when are they identical?

Concept Question 3-2: When is the position vector of a point identical to the distance vector between two points?

Concept Question 3-3: If $\mathbf{A}\cdot\mathbf{B}=0$, what is θ_{AB}?

Concept Question 3-4: If $\mathbf{A}\times\mathbf{B}=0$, what is θ_{AB}?

Concept Question 3-5: Is $\mathbf{A}(\mathbf{B}\cdot\mathbf{C})$ a vector triple product?

Concept Question 3-6: If $\mathbf{A}\cdot\mathbf{B}=\mathbf{A}\cdot\mathbf{C}$, does it follow that $\mathbf{B}=\mathbf{C}$?

3-2 Orthogonal Coordinate Systems

A three-dimensional coordinate system allows us to uniquely specify locations of points in space and the magnitudes and directions of vectors. Coordinate systems may be orthogonal or nonorthogonal.

► An ***orthogonal coordinate system*** is one in which coordinates are measured along locally mutually perpendicular axes. ◄

Nonorthogonal systems are very specialized and seldom used in solving practical problems. Many orthogonal coordinate systems have been devised, but the most commonly used are

- the Cartesian (also called rectangular),
- the cylindrical, and
- the spherical coordinate system.

Why do we need more than one coordinate system? Whereas a point in space has the same location and an object has the same shape regardless of which coordinate system is used to describe them, the solution of a practical problem can be greatly facilitated by the choice of a coordinate system that best fits the geometry under consideration. The following subsections examine the properties of each of the aforementioned orthogonal systems, and Section 3-3 describes how a point or vector may be transformed from one system to another.

3-2.1 Cartesian Coordinates

The Cartesian coordinate system was introduced in Section 3-1 to illustrate the laws of vector algebra. Instead of repeating these laws for the Cartesian system, we summarize them in **Table 3-1**. Differential calculus involves the use of differential lengths, areas, and volumes. In Cartesian coordinates, a ***differential length vector*** (**Fig. 3-8**) is expressed as

$$d\mathbf{l}=\hat{\mathbf{x}}\,dl_x+\hat{\mathbf{y}}\,dl_y+\hat{\mathbf{z}}\,dl_z=\hat{\mathbf{x}}\,dx+\hat{\mathbf{y}}\,dy+\hat{\mathbf{z}}\,dz, \qquad (3.34)$$

where $dl_x=dx$ is a differential length along $\hat{\mathbf{x}}$, and similar interpretations apply to $dl_y=dy$ and $dl_z=dz$.

A ***differential area vector*** $d\mathbf{s}$ is a vector with magnitude ds equal to the product of two differential lengths (such as dl_y and dl_z) and direction specified by a unit vector along the third

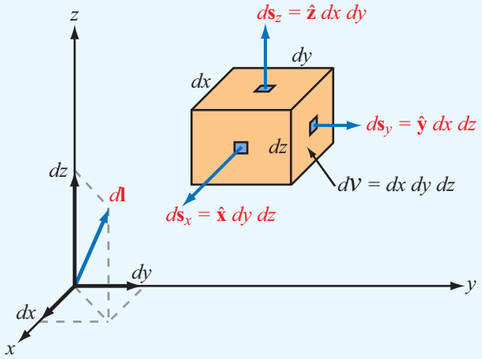


Figure 3-8 Differential length, area, and volume in Cartesian coordinates.

和

$$(\mathbf{A}\times\mathbf{B})\times\mathbf{C}=\begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ -3 & -1 & 1\\ -2 & 0 & 3\end{vmatrix}=-\hat{\mathbf{x}}3+\hat{\mathbf{y}}7-\hat{\mathbf{z}}2.$$

类似的过程给出 A × (B × C) = x̂2 + ŷ4 + ẑ。两个向量三元乘积的结果不同这一事实证明了等式中所述的不等式。 (3.32) 。

Concept Question 3-1: When are two vectors equal and when are they identical?

Concept Question 3-2: When is the position vector of a point identical to the distance vector between two points?

Concept Question 3-3: If $\mathbf{A}\cdot\mathbf{B}=0$, what is θ_{AB}?

Concept Question 3-4: If $\mathbf{A}\times\mathbf{B}=0$, what is θ_{AB}?

Concept Question 3-5: Is $\mathbf{A}(\mathbf{B}\cdot\mathbf{C})$ a vector triple product?

Concept Question 3-6: If $\mathbf{A}\cdot\mathbf{B}=\mathbf{A}\cdot\mathbf{C}$, does it follow that $\mathbf{B}=\mathbf{C}$?

3-2 正交坐标系

三维坐标系允许我们唯一地指定空间中点的位置以及向量的大小和方向。坐标系可以是正交的或非正交的。

► 正交坐标系是一种沿局部相互垂直的轴测量坐标的坐标系。 ◄

非正交系统非常专业，很少用于解决实际问题。已经设计了许多正交坐标系，但最常用的是

- 笛卡尔（也称为矩形），
- 圆柱形，并且
- 球坐标系。

为什么我们需要多个坐标系？尽管无论使用哪种坐标系来描述，空间中的点都具有相同的位置，物体也具有相同的形状，但通过选择最适合所考虑的几何形状的坐标系，可以极大地促进实际问题的解决。以下小节检查了上述每个正交系统的属性，第 3-3 节描述了如何将点或向量从一个系统变换到另一个系统。

3-2.1 笛卡尔坐标

第 3-1 节介绍了笛卡尔坐标系，以说明向量代数的定律。我们没有在笛卡尔系统中重复这些定律，而是将它们总结在表 3-1 中。微分学涉及不同长度、面积和体积的使用。在笛卡尔坐标系中，微分长度向量（图3-8）表示为

$$d\mathbf{l}=\hat{\mathbf{x}}\,dl_x+\hat{\mathbf{y}}\,dl_y+\hat{\mathbf{z}}\,dl_z=\hat{\mathbf{x}}\,dx+\hat{\mathbf{y}}\,dy+\hat{\mathbf{z}}\,dz, \qquad (3.34)$$

其中 $dl_x = dx$ 是沿 x̂ 的微分长度，类似的解释适用于 $dl_y = dy$ 和 $dl_z = dz$。

微分面积向量 ds 是一个向量，其大小 ds 等于两个微分长度（例如 dl_y 和 dl_z）和由沿第三个单位向量指定的方向的乘积。

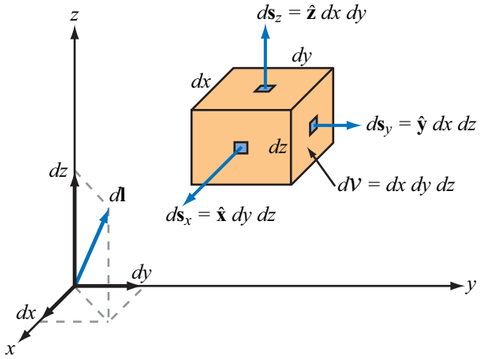


图 3-8 笛卡尔坐标系中的长度、面积和体积微分。

Table 3-1 Summary of vector relations.

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $\lvert\mathbf{A}\rvert =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1$, for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1$, for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1$, for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}}\times\hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}}\cdot\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}}\times\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A}\cdot\mathbf{B} =$	$A_xB_x + A_yB_y + A_zB_z$	$A_rB_r + A_\phi B_\phi + A_zB_z$	$A_RB_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A}\times\mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{r}}\,dr + \hat{\boldsymbol{\phi}}r\,d\phi + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{R}}\,dR + \hat{\boldsymbol{\theta}}R\,d\theta + \hat{\boldsymbol{\phi}}R\sin\theta\,d\phi$
Differential surface areas	$d\mathbf{s}_x = \hat{\mathbf{x}}\,dy\,dz$ $d\mathbf{s}_y = \hat{\mathbf{y}}\,dx\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}\,dx\,dy$	$d\mathbf{s}_r = \hat{\mathbf{r}}r\,d\phi\,dz$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dr\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}r\,dr\,d\phi$	$d\mathbf{s}_R = \hat{\mathbf{R}}R^2\sin\theta\,d\theta\,d\phi$ $d\mathbf{s}_\theta = \hat{\boldsymbol{\theta}}R\sin\theta\,dR\,d\phi$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}R\,dR\,d\theta$
Differential volume $d\upsilon =$	$dx\,dy\,dz$	$r\,dr\,d\phi\,dz$	$R^2\sin\theta\,dR\,d\theta\,d\phi$

direction (such as $\hat{\mathbf{x}}$). Thus, for a differential area vector in the y–z plane,

$$d\mathbf{s}_x = \hat{\mathbf{x}}\,dl_y\,dl_z = \hat{\mathbf{x}}\,dy\,dz \qquad (y\text{–}z \text{ plane}), \tag{3.35a}$$

with the subscript on $d\mathbf{s}$ denoting its direction. Similarly,

$$d\mathbf{s}_y = \hat{\mathbf{y}}\,dx\,dz \qquad (x\text{–}z \text{ plane}), \tag{3.35b}$$

$$d\mathbf{s}_z = \hat{\mathbf{z}}\,dx\,dy \qquad (x\text{–}y \text{ plane}). \tag{3.35c}$$

A ***differential volume*** equals the product of all three differential lengths:

$$d\upsilon = dx\,dy\,dz. \tag{3.36}$$

3-2.2 Cylindrical Coordinates

The cylindrical coordinate system is useful for solving problems involving structures with cylindrical symmetry, such as calculating the capacitance per unit length of a coaxial transmission line. In the cylindrical coordinate system, the location of a point in space is defined by three variables: r, ϕ, and z (**Fig. 3-9**). The coordinate r is the ***radial distance*** in the x–y the ***azimuth angle*** measured from the positive x axis, and z is as previously defined in the Cartesian coordinate system. Their ranges are $0 \le r < \infty$, $0 \le \phi < 2\pi$, and $-\infty < z < \infty$. Point $P(r_1, \phi_1, z_1)$ in **Fig. 3-9** is located at the intersection of three surfaces. These are the cylindrical surface defined by $r = r_1$, the vertical half-plane defined by $\phi = \phi_1$ (which extends outwardly from the z axis), and the horizontal plane defined by $z = z_1$.

▶ The mutually perpendicular base vectors are $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ with $\hat{\mathbf{r}}$ pointing away from the origin along r, $\hat{\boldsymbol{\phi}}$ pointing in a direction tangential to the cylindrical surface, and $\hat{\mathbf{z}}$ pointing along the vertical. Unlike the Cartesian system, where base vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ are independent of the location of P, both $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ are functions of ϕ in the cylindrical system. ◀

138 CHAPTER 3 VECTOR ANALYSIS

表 3-1 向量关系汇总。

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $\lvert\mathbf{A}\rvert =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1$, for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1$, for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1$, for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}}\cdot\hat{\mathbf{y}} = \hat{\mathbf{y}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}}\times\hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}}\times\hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{z}} = \hat{\mathbf{z}}\cdot\hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}}\times\hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}}\cdot\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}\cdot\hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}}\cdot\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}\cdot\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}}\times\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}}\times\hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}}\times\hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A}\cdot\mathbf{B} =$	$A_xB_x + A_yB_y + A_zB_z$	$A_rB_r + A_\phi B_\phi + A_zB_z$	$A_RB_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A}\times\mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{r}}\,dr + \hat{\boldsymbol{\phi}}r\,d\phi + \hat{\mathbf{z}}\,dz$	$\hat{\mathbf{R}}\,dR + \hat{\boldsymbol{\theta}}R\,d\theta + \hat{\boldsymbol{\phi}}R\sin\theta\,d\phi$
Differential surface areas	$d\mathbf{s}_x = \hat{\mathbf{x}}\,dy\,dz$ $d\mathbf{s}_y = \hat{\mathbf{y}}\,dx\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}\,dx\,dy$	$d\mathbf{s}_r = \hat{\mathbf{r}}r\,d\phi\,dz$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dr\,dz$ $d\mathbf{s}_z = \hat{\mathbf{z}}r\,dr\,d\phi$	$d\mathbf{s}_R = \hat{\mathbf{R}}R^2\sin\theta\,d\theta\,d\phi$ $d\mathbf{s}_\theta = \hat{\boldsymbol{\theta}}R\sin\theta\,dR\,d\phi$ $d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}R\,dR\,d\theta$
Differential volume $d\upsilon =$	$dx\,dy\,dz$	$r\,dr\,d\phi\,dz$	$R^2\sin\theta\,dR\,d\theta\,d\phi$

方向（例如 x̂）。因此，对于 y-z 平面上的微分面积矢量，

$$d\mathbf{s}_x = \hat{\mathbf{x}}\,dl_y\,dl_z = \hat{\mathbf{x}}\,dy\,dz \text{（y–z 平面）}, \tag{3.35a}$$

ds 上的下标表示其方向。相似地，

$$d\mathbf{s}_y = \hat{\mathbf{y}}\,dx\,dz \qquad (x\text{–}z \text{ plane}), \tag{3.35b}$$

$$d\mathbf{s}_z = \hat{\mathbf{z}}\,dx\,dy \qquad (x\text{–}y \text{ plane}). \tag{3.35c}$$

微分体积等于所有三个微分长度的乘积：

$$d\upsilon = dx\,dy\,dz. \tag{3.36}$$

3-2.2 圆柱坐标

柱坐标系可用于解决涉及柱对称结构的问题，例如计算同轴传输线单位长度的电容。在柱坐标系中，空间中点的位置由三个变量定义：r、ϕ、z（图3-9）。坐标 r 是 x-y 方向上的径向距离，即从正 x 轴测量的方位角，z 是先前在笛卡尔坐标系中定义的。它们的范围是 $0 \le r < \infty$、$0 \le \phi < 2\pi$ 和 $-\infty < z < \infty$。图3-9中的点$P(r_1,\phi_1,z_1)$位于三个曲面的交点处。它们是由 $r = r_1$ 定义的圆柱面，由从 z 轴）向外延伸的 $\phi = \phi_1$（定义的垂直半平面，以及由 $z = z_1$ 定义的水平面。

► 相互垂直的基向量为 r̂、$\hat{\boldsymbol{\phi}}$ 和 ẑ，其中 r̂ 沿 r 远离原点，$\hat{\boldsymbol{\phi}}$ 指向与圆柱面相切的方向，ẑ 沿垂直方向指向。与笛卡尔坐标系不同，在笛卡尔坐标系中，基向量 x̂、ŷ 和 ẑ 与 P 的位置无关，而 r̂ 和 $\hat{\boldsymbol{\phi}}$ 都是圆柱坐标系中 ϕ 的函数。◄

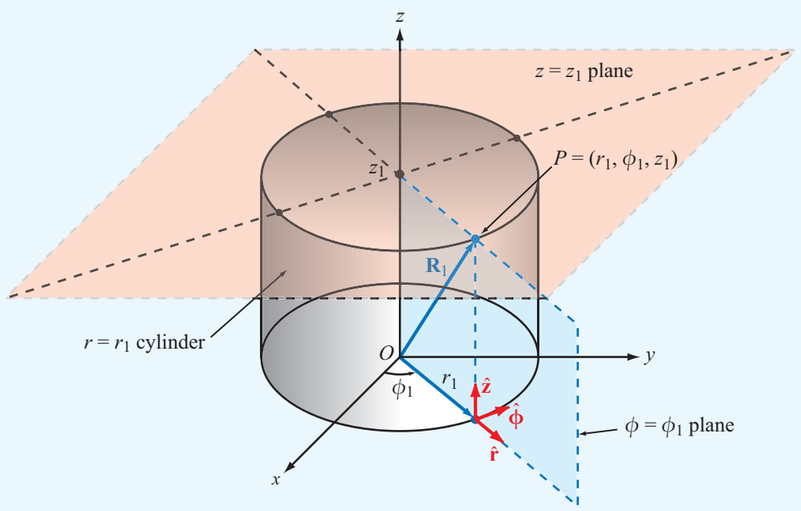


Figure 3-9 Point $P(r_1, \phi_1, z_1)$ in cylindrical coordinates; r_1 is the radial distance from the origin in the x–y plane, ϕ_1 is the angle in the x–y plane measured from the x axis toward the y axis, and z_1 is the vertical distance from the x–y plane.

The base unit vectors obey the following right-hand cyclic relations:

$$\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}, \qquad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}, \qquad \hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}, \tag{3.37}$$

and like all unit vectors, $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$, and $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$.

In cylindrical coordinates, a vector is expressed as

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z, \tag{3.38}$$

where A_r, A_ϕ, and A_z are the components of $\mathbf{A}$ along the $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ directions. The magnitude of $\mathbf{A}$ is obtained by applying Eq. (3.17), which gives

$$|\mathbf{A}| = \sqrt[+]{\mathbf{A} \cdot \mathbf{A}} = \sqrt[+]{A_r^2 + A_\phi^2 + A_z^2}. \tag{3.39}$$

The position vector $\overrightarrow{OP}$ shown in **Fig. 3-9** has components along r and z only. Thus,

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1. \tag{3.40}$$

The dependence of $\mathbf{R}_1$ on ϕ_1 is implicit through the dependence of $\hat{\mathbf{r}}$ on ϕ_1. Hence, when using Eq. (3.40) to denote the position vector of point $P(r_1, \phi_1, z_1)$, it is necessary to specify that $\hat{\mathbf{r}}$ is at ϕ_1.

Figure 3-10 shows a differential volume element in cylindrical coordinates. The differential lengths along $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$ are

$$dl_r = dr, \qquad dl_\phi = r\,d\phi, \qquad dl_z = dz. \tag{3.41}$$

Note that the differential length along $\hat{\boldsymbol{\phi}}$ is $r\,d\phi$, not just $d\phi$. The differential length $d\mathbf{l}$ in cylindrical coordinates is given by

$$d\mathbf{l} = \hat{\mathbf{r}}\,dl_r + \hat{\boldsymbol{\phi}}\,dl_\phi + \hat{\mathbf{z}}\,dl_z = \hat{\mathbf{r}}\,dr + \hat{\boldsymbol{\phi}}r\,d\phi + \hat{\mathbf{z}}\,dz. \tag{3.42}$$

As was stated previously for the Cartesian coordinate system, the product of any pair of differential lengths is equal to the magnitude of a vector differential surface area with a surface normal pointing along the direction of the third coordinate. Thus,

$$d\mathbf{s}_r = \hat{\mathbf{r}}\,dl_\phi\,dl_z = \hat{\mathbf{r}}r\,d\phi\,dz \quad \textbf{(ϕ–z cylindrical surface)}, \tag{3.43a}$$

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dl_r\,dl_z = \hat{\boldsymbol{\phi}}\,dr\,dz \quad \textbf{(r–z plane)}, \tag{3.43b}$$

$$d\mathbf{s}_z = \hat{\mathbf{z}}\,dl_r\,dl_\phi = \hat{\mathbf{z}}r\,dr\,d\phi \quad \textbf{(r–ϕ plane)}. \tag{3.43c}$$

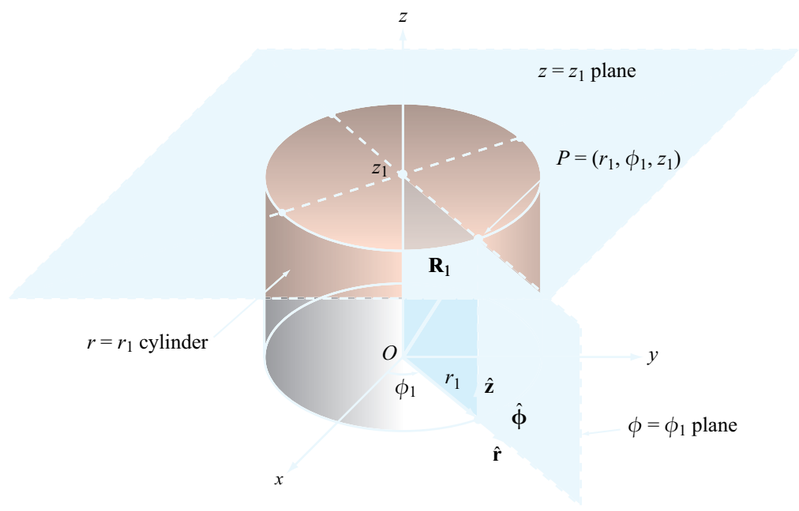


图3-9 柱坐标中的点$P(r_1,\phi_1,z_1)$； r_1 是 x-y 平面中距原点的径向距离，ϕ_1 是 x-y 平面中从 x 轴朝向 y 轴测量的角度，z_1 是距 x-y 平面的垂直距离。

基本单位向量遵循以下右手循环关系：

$$\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}, \qquad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}, \qquad \hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}, \tag{3.37}$$

与所有单位向量一样，$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ 和 $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$。

在柱坐标系中，向量表示为

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z, \tag{3.38}$$

其中 A_r、A_ϕ 和 A_z 是 A 沿 $\hat{\mathbf{r}}$、$\hat{\boldsymbol{\phi}}$ 和 $\hat{\mathbf{z}}$ 方向的分量。 A 的大小可通过应用方程式获得。 (3.17)，给出

$$|\mathbf{A}| = \sqrt[+]{\mathbf{A} \cdot \mathbf{A}} = \sqrt[+]{A_r^2 + A_\phi^2 + A_z^2}. \tag{3.39}$$

图 3-9 所示的位置向量 $\overrightarrow{OP}$ 仅具有沿 r 和 z 的分量。因此，

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1. \tag{3.40}$$

R_1 对 ϕ_1 的依赖是通过 $\hat{\mathbf{r}}$ 对 ϕ_1 的依赖来隐含的。因此，当使用方程式时。 (3.40) 表示点 $P(r_1, \phi_1, z_1)$ 的位置向量，需要指定 $\hat{\mathbf{r}}$ 在 ϕ_1 处。

图 3-10 显示了柱坐标系中的微分体积单元。沿 $\hat{\mathbf{r}}$、$\hat{\boldsymbol{\phi}}$ 和 $\hat{\mathbf{z}}$ 的差分长度为

$$dl_r = dr, \qquad dl_\phi = r\,d\phi, \qquad dl_z = dz. \tag{3.41}$$

请注意，沿 $\hat{\boldsymbol{\phi}}$ 的差分长度是 $r\,d\phi$，而不仅仅是 $d\phi$。柱坐标系中的微分长度 dl 由下式给出

$$d\mathbf{l} = \hat{\mathbf{r}}\,dl_r + \hat{\boldsymbol{\phi}}\,dl_\phi + \hat{\mathbf{z}}\,dl_z = \hat{\mathbf{r}}\,dr + \hat{\boldsymbol{\phi}}r\,d\phi + \hat{\mathbf{z}}\,dz. \tag{3.42}$$

正如前面针对笛卡尔坐标系所述，任何一对微分长度的乘积都等于向量微分表面积的大小，其中表面法线指向第三坐标的方向。因此，

$$d\mathbf{s}_r = \hat{\mathbf{r}}\,dl_\phi\,dl_z = \hat{\mathbf{r}}r\,d\phi\,dz \quad \textbf{(ϕ–z cylindrical surface)}, \tag{3.43a}$$

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dl_r\,dl_z = \hat{\boldsymbol{\phi}}\,dr\,dz \quad \textbf{(r–z plane)}, \tag{3.43b}$$

$$d\mathbf{s}_z = \hat{\mathbf{z}}\,dl_r\,dl_\phi = \hat{\mathbf{z}}r\,dr\,d\phi \quad \textbf{(r–ϕ plane)}. \tag{3.43c}$$

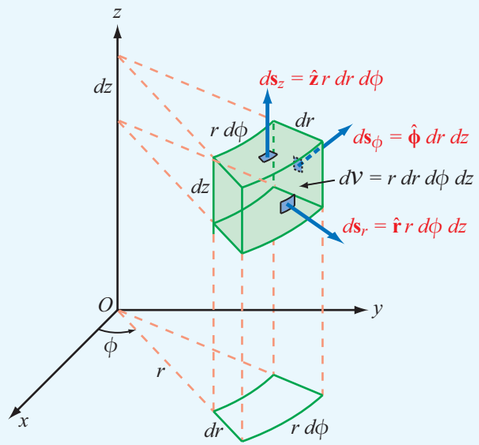


Figure 3-10 Differential areas and volume in cylindrical coordinates.

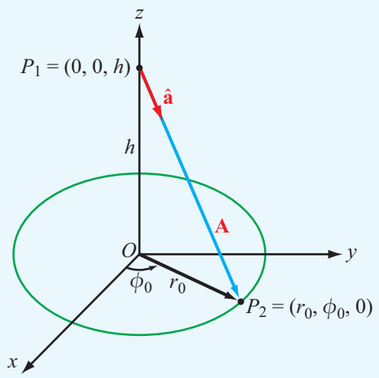


Figure 3-11 Geometry of Example 3-4.

The differential volume is the product of the three differential lengths,

$$dv = dl_r\, dl_\phi\, dl_z = r\, dr\, d\phi\, dz. \tag{3.44}$$

These properties of the cylindrical coordinate system are summarized in **Table 3-1**.

Example 3-4: Distance Vector in Cylindrical Coordinates

Find an expression for the unit vector of vector **A** shown in **Fig. 3-11** in cylindrical coordinates.

Solution: In triangle OP_1P_2,

$$\overrightarrow{OP_2} = \overrightarrow{OP_1} + \mathbf{A}.$$

Hence,

$$\mathbf{A} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \hat{\mathbf{r}} r_0 - \hat{\mathbf{z}} h,$$

and

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{\mathbf{r}} r_0 - \hat{\mathbf{z}} h}{\sqrt{r_0^2 + h^2}}.$$

We note that the expression for **A** is independent of ϕ_0. This implies that all vectors from point P_1 to any point on the circle defined by $r = r_0$ in the x–y plane are equal in the cylindrical coordinate system, which is not true. The ambiguity can be resolved by specifying that **A** passes through a point whose $\phi = \phi_0$.

Example 3-5: Cylindrical Area

Find the area of a cylindrical surface described by $r = 5$, $30° \le \phi \le 60°$, and $0 \le z \le 3$ (**Fig. 3-12**).

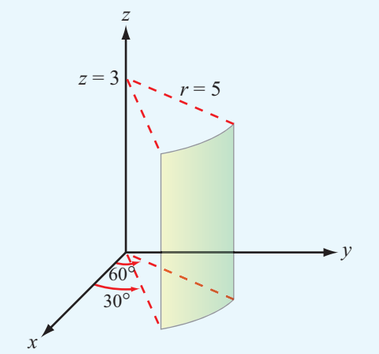


Figure 3-12 Cylindrical surface of Example 3-5.

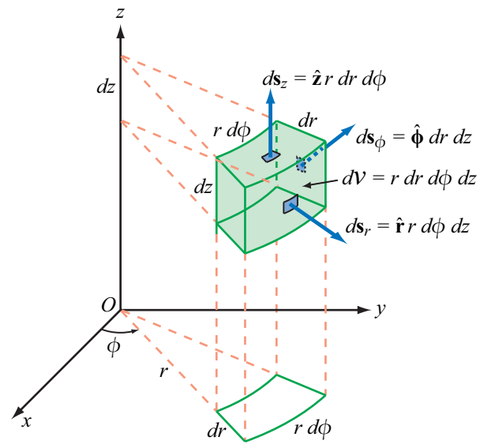


图 3-10 柱坐标系中的微分面积和体积。

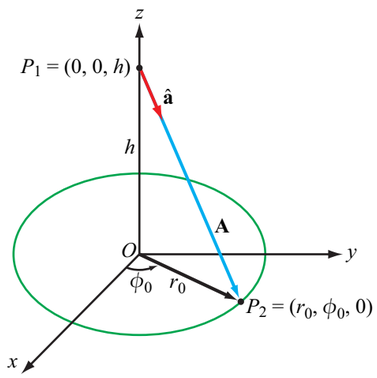


图 3-11 示例 3-4 的几何形状。

微分体积是三个微分长度的乘积，

$$dv = dl_r\, dl_\phi\, dl_z = r\, dr\, d\phi\, dz. \tag{3.44}$$

表 3-1 总结了柱坐标系的这些属性。

示例 3-4：柱坐标系中的距离向量

求图 3-11 所示矢量 A 的单位矢量在柱坐标系中的表达式。

解：在三角形OP_1P_2中，

$$\overrightarrow{OP_2} = \overrightarrow{OP_1} + \mathbf{A}.$$

因此，

$$\mathbf{A} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \hat{\mathbf{r}} r_0 - \hat{\mathbf{z}} h,$$

和

$$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{\mathbf{r}} r_0 - \hat{\mathbf{z}} h}{\sqrt{r_0^2 + h^2}}.$$

我们注意到 A 的表达式独立于 ϕ_0。这意味着从点 P_1 到 x-y 平面中由 $r = r_0$ 定义的圆上的任何点的所有向量在圆柱中都相等

坐标系，这是不正确的。可以通过指定 A 经过其 $\phi = \phi_0$ 的点来解决歧义。

例 3-5：圆柱面积

求 $r = 5$、$30° \le \phi \le 60°$ 和 $0 \le z \le 3$ 描述的圆柱表面的面积（图 3-12）。

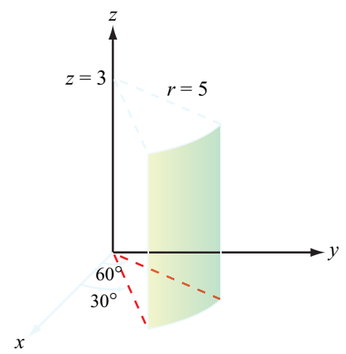


图3-12 例3-5的圆柱面。